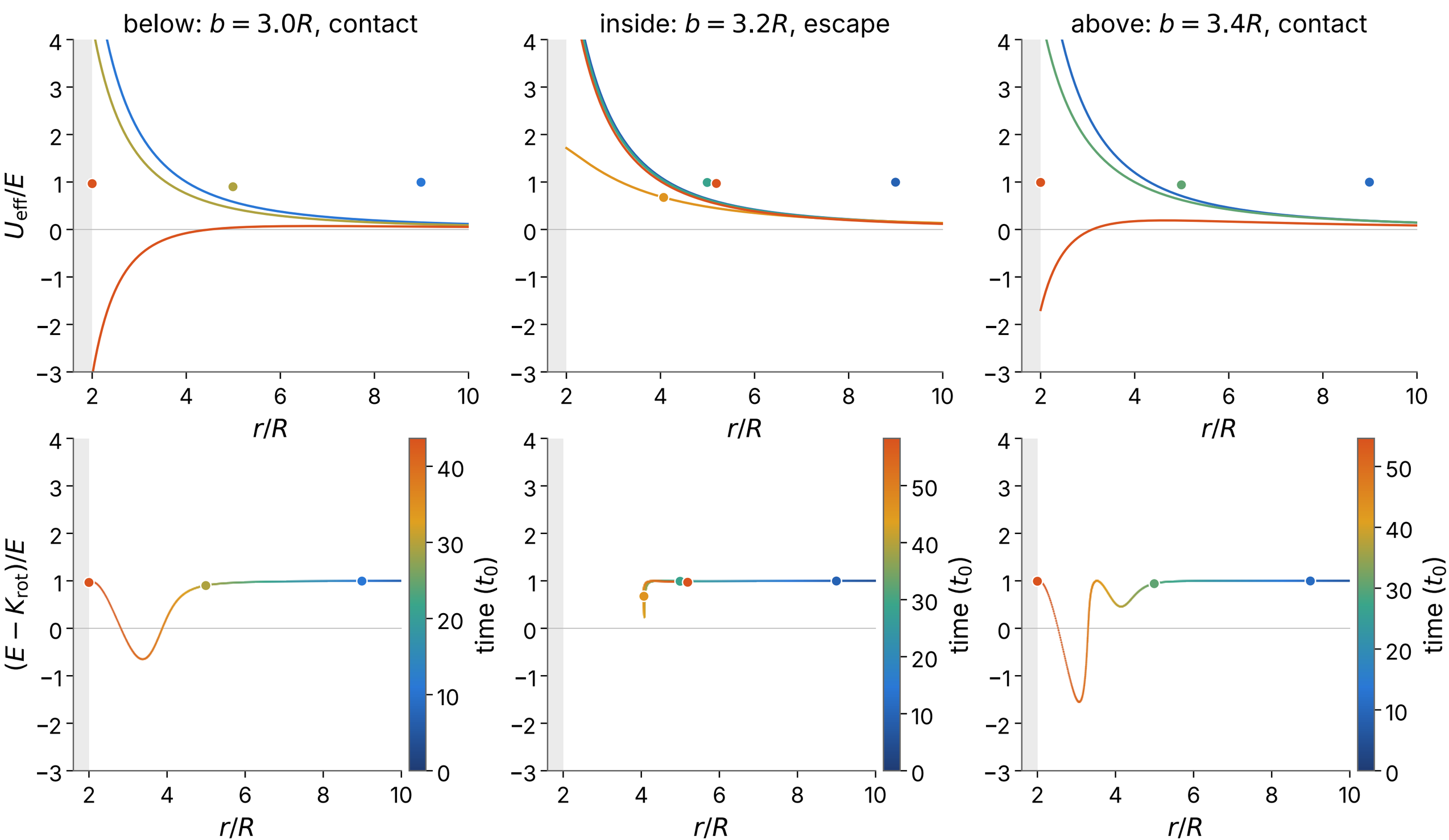

below: $b = 3.0R$, contact
inside: $b = 3.2R$, escape
above: $b = 3.4R$, contact
U_{eff}/E
$(E - K_{\mathrm{rot}})/E$
r/R
time (t_0)